# Isaac Oliva-González

isaac.oliva23@gmail.com Scholar 0009-0006-2174-8861 IOlivaG isaacoliva23 Personal Site

## Summary

Graduate researcher in computer science with a background in chemical engineering. My research focuses on the intersection of mathematical optimisation, operations research, and quantum computing, with an emphasis on modelling and analysing structured optimisation problems. I have worked on combinatorial optimisation and mathematical programming models, as well as on symmetry and algebraic techniques to improve computational performance. My interests include algorithmic development for large-scale problems arising in energy and process systems.

## Research Interests

**Optimization & Operations Research**
Combinatorial Optimization, Mathematical Programming, Algebraic Reformulations, Graph-Theoretic Approaches

**Quantum Computing & Algorithms**
Hybrid Quantum-Classical Methods, Quantum Machine Learning, Quantum Annealing

**Applications**
Energy Systems, Process Systems Engineering, Large-Scale Optimisation

## Education

**MSc in Computer Science** *Aug 2023 - Aug 2025*
Universidad Autónoma de Querétaro Querétaro, MEX

- Dissertation on quantum annealing and simulated annealing for discrete-space route optimization.

**BSc in Chemical Engineering** *Jan 2017 - Dec 2021*
Universidad de Guanajuato Guanajuato, MEX

- Dissertation on modeling and design of intensified distillation sequences for renewable fuel separation.

## Experience

**Tecnológico de Monterrey** *Jan 2025 - Aug 2025*
Visiting Research Scholar | *Supervisor: Dr. Ilse María Hernández-Romero* Monterrey, MEX

- Conducted collaborative research on hybrid quantum–classical neural architectures using variational quantum circuits to model black carbon dynamics, benchmarking against classical models to assess the contribution of quantum-enhanced architectures in predictive performance and model expressiveness (*PennyLane*). See [S1].

**Universidad Autónoma de Querétaro** *Aug 2023 - Aug 2025*
MSc Student Querétaro, MEX

- Designed and benchmarked quantum and classical optimisation algorithms for discrete shortest-path problems, analysing performance trade-offs across solution quality and computational efficiency (*Julia*). See [S2].
- Designed and implemented finite automata for encoding alphabetic sequences and performing operations on deterministic finite automata (*Mathematica*).

**Ingredion México** *Feb 2022 - Dec 2022*
Process Technician San Juan del Río, MEX

- Operated and monitored industrial starch processing systems, performing on-site troubleshooting and real-time adjustments to ensure process reliability.

**NOVARES México** *Jan 2021 - Jun 2021*
Undergraduate Intern Silao, MEX

- Supported the implementation of health, safety, and environmental (HSE) programs in an automotive manufacturing environment.

**Instituto Tecnológico de Villahermosa** *Summer 2019*
Visiting Research Scholar | *Supervisor: Dr. Francisco López-Villarreal* Villahermosa, MEX

- Developed a computational interface for vapor–liquid equilibrium calculations in isothermal flash distillation, applying numerical methods to thermodynamic optimization problems (*MATLAB*).

## Research Projects

**Symmetry Reduction in Combinatorial Optimization** *2026*
Available here

- Exploratory study of symmetry reduction in combinatorial optimisation (ILP), including shortest path, graph coloring, max-cut, and TSP. Applied lexicographic constraints, variable fixing, and automorphism-based methods to reduce redundant solutions (*Julia, JuMP, HiGHS*).

**Symmetry Reduction in Combined Cycle Unit Commitment** *2026*
Available here

- Study of symmetry reduction applied to the Combined Cycle Min-Up/Min-Down Unit Commitment Problem (CC-MUCP). Derived demand-aware lexicographic constraints to break the two-level permutation symmetry arising from identical packages and gas turbines (*Julia, JuMP, HiGHS*).

## Skills

**Programming**: Python, Julia
**Optimisation & Modelling**: JuMP, GAMS, MATLAB
**Quantum & Scientific Computing**: PennyLane, Qiskit, Mathematica, Aspen Plus, Jupyter Notebooks

## Languages

**Spanish**: Native proficiency.
**English**: Proficient (TOEFL ITP 562 | CEFR B2).
**German**: Elementary proficiency (CEFR A2).

## Research & Publications

*Book Chapters*

[B1] I. M. Hernández-Romero, **I. Oliva-González**, O. D. Lara-Montaño, K. Hernández-Romero, G. G. Ezquivel-Patiño, O. Kharissova, A. I. Ramirez, L. T. González, "Integrating Quantum Computing into Sustainable Carbon-Capture Materials Research: Opportunities and Perspectives", *Handbook of Carbon Negative Footprint Materials*, Springer Nature (*accepted*), 2026.

*Preprints & Manuscripts in Progress*

[S1] **I. Oliva-González**, L. T. González, O. D. Lara-Montaño, A. I. Ramirez, A. Mendoza, and I. M. Hernández-Romero, "Quantum-enhanced and classical machine learning for black carbon forecasting integrating meteorological parameters and criteria pollutants," *EPJ Quantum Technology* (*under review*), 2026.

[S2] **I. Oliva-González** and H. Jiménez-Hernández, "QUBO-based simulated annealing approach for the shortest path problem with applications to urban transportation networks," *Modelling* (*under review*), 2026. Preprint available, DOI: 10.20944/preprints202511.0578.v1

*Conference Proceedings*

[C1] **I. Oliva-González**, A. G. Romero-Izquierdo, C. Gutiérrez-Antonio, F. I. Gómez-Castro, and S. Hernández, "Síntesis de columnas con múltiples paredes divisorias para la separación de combustible sustentable de aviación," *Avances en Ingeniería Química*, vol. 3, no. 1, Memorias del XLV Encuentro Nacional de la AMIDIQ – Ingeniería de Procesos, Simulación y Control, México, 2024, pp. 265–270. (ISSN 2683-2925). Available here.

[C2] **I. Oliva-González**, A. G. Romero-Izquierdo, C. Gutiérrez-Antonio, F. I. Gómez-Castro, and S. Hernández, "Computer-aided design of intensified separation sequences for a complex mixture of renewable hydrocarbons," *Proceedings of the 34th European Symposium on Computer Aided Process Engineering/15th International Symposium of Process Systems Engineering* (PSE2024/ESCAPE34), Elsevier, vol. 53, 2024, pp. 1483–1488. DOI:10.1016/B978-0-443-28824-1.50248-9

[C3] **I. Oliva-González**, C. Gutiérrez-Antonio, F. I. Gómez-Castro, S. Hernández, and A. G. Romero-Izquierdo, "Síntesis de columnas con múltiples paredes divisorias para la separación de combustible sustentable de aviación," *Avances en Ingeniería Química*, vol. 2, no. 1, Memorias del XLIV Encuentro Nacional de la AMIDIQ – Ingeniería de Procesos, Simulación y Control, México, 2023, pp. 402–407. (ISSN 2683-2925). Available here.

[C4] **I. Oliva-González**, C. Gutiérrez-Antonio, F. I. Gómez-Castro, E. Quiroz Pérez, S. Hernández, and A. G. Romero-Izquierdo, "Síntesis de secuencias intensificadas de destilación para la separación de una mezcla multicomponente de hidrocarburos renovables," *Avances en Ingeniería Química*, vol. 1, no. 4, Memorias del XLIII Encuentro Nacional de la AMIDIQ – Ingeniería de Procesos, Simulación y Control, México, 2022, pp. 218–223. (ISSN 2683-2925). Available here.

*Conference Presentations*

[P1] **I. Oliva-González**, A. G. Romero-Izquierdo, C. Gutiérrez-Antonio, F. I. Gómez-Castro, and S. Hernández, "Síntesis de columnas con múltiples paredes divisorias para la separación de combustible sustentable de aviación," in *Retos y Oportunidades al Net-Zero 2030. Encuentro PSPB-FI01*, Querétaro, México, 2023. *(Oral presentation)*

[P2] **I. Oliva-González**, A. G. Romero-Izquierdo, C. Gutiérrez-Antonio, F. I. Gómez-Castro, and S. Hernández, "Synthesis of alternative intensified sequences for the separation of a multicomponent renewable hydrocarbons mixture," in *12th International Conference Distillation & Absorption 2022* (DA2022), Touluse, France, 2022. *(Poster presentation)*